

Non-Locality and the Boundary of Reality: Quantum Potential, Irreversible Actualization, and the Thermodynamic Origin of Classical Existence

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Abstract

Quantum non-locality — the apparent ability of entangled systems to exhibit correlated behavior regardless of spatial separation — has been called the defining mystery of quantum mechanics. We argue that it is not mysterious at all, but is the expected behavior of physical systems that have not yet crossed the boundary between potential and actual. Within the Informational Actualization Model (IAM), that boundary has a precise thermodynamic identity: every irreversible quantum-to-classical transition pays a cost of $k_B T_H \ln 2$ per bit to the cosmic horizon, where $T_H = \hbar H / 2\pi k_B$ is the Gibbons–Hawking temperature. The crossing is physically real, thermodynamically irreversible, and cosmologically consequential.

Before this crossing, a quantum system is not a particle with hidden properties. It is a field of potential — a superposition that has not yet been written to the ledger of classical existence. Non-locality is the expected property of potential: potential has no position, no trajectory, no classical identity. After the crossing, the system becomes a classical particle, the information is encoded permanently on the cosmic horizon, and locality is restored. There is no spooky action at a distance because there is no particle to act at a distance until actualization occurs.

This identification — that the boundary of classical existence is a thermodynamic threshold, not a mathematical postulate — is a consequence of the Jacobson–Cai–Kim derivation extended to include the Landauer cost of irreversible bulk information production. The accumulated ledger of 13.8 billion years of such crossings is measured by the activation function $\mathcal{E}(a) = \exp(1 - 1/a)$, derived from horizon thermodynamics. Its cosmological consequences — $\mu(a) < 1$ for matter on timelike worldlines, $\Sigma(a) = 1$ for photons on null geodesics — are confirmed against Planck 2018 through 17 converged MCMC chains with no free parameters beyond Λ CDM. The bridge from the quantum boundary to the cosmological scale is the critical exponent $n = 5/2$, derived independently from quantum decoherence physics (bottom-up) and from

horizon thermodynamic consistency (top-down). Their convergence constitutes a derivation, not a scaling argument.

The predicted matter-sector modification today is $\mu_0 - 1 = -0.136$, derived from first principles with zero free parameters.

1 The Mystery and the Category Error

Quantum non-locality has been presented as one of the deepest mysteries in physics. Entangled particles, separated by arbitrary distances, exhibit correlated measurement outcomes that cannot be explained by any local hidden variable theory [Bell, 1964]. The correlations violate Bell inequalities [Aspect et al., 1982], the violation has been confirmed to high confidence [Hensen et al., 2015], and the conclusion seems inescapable: nature is non-local.

We propose that this conclusion rests on a category error. The mystery of non-locality arises from applying the language of classical particles — position, trajectory, definite state — to physical systems that are not yet classical particles. Before a quantum-to-classical transition occurs, there is no particle. There is a quantum field in superposition: a distribution of potential that has not paid the thermodynamic cost of becoming actual. Asking where the entangled “particle” is, or how it communicates with its partner, is like asking where a wave is before it breaks. The question assumes a classical object that does not yet exist.

Within the Informational Actualization Model (IAM) [Mahaffey, 2026e,d], this distinction is not philosophical. It is derived. The boundary between potential and actual has a precise thermodynamic identity: every irreversible quantum-to-classical transition pays a Landauer cost [Landauer, 1961] of $k_B T_H \ln 2$ per bit to the cosmic horizon. Before payment, the system is potential. After payment, it is classical. The boundary is real, irreversible, and cosmologically consequential.

2 The Ontological Distinction: Potential and Actual

Standard quantum mechanics describes two regimes that are treated as limits of a single formalism: the quantum regime of superposition and entanglement, and the classical regime of definite states and particles. The transition between them is handled by the measurement postulate — a postulate, not a derivation. IAM derives it.

The distinction is ontological, not merely mathematical. A quantum system in superposition is not a classical particle with unknown properties. It is categorically different: a field of potential governed by the Schrödinger equation, with no definite position, no definite momentum, and no classical identity. The word “particle” already assumes actualization. Applied to a pre-measurement quantum system, it is a category error that has generated a century of apparent paradoxes.

IAM identifies the crossing from potential to actual with a specific physical event: an irreversible thermodynamic interaction between a quantum system and matter on a timelike worldline that dissipates energy $Q \geq k_B T \ln 2$ — the Landauer threshold [Landauer, 1961]. This is not a postulate. It follows from requiring that the entropy budget of the Jacobson–Cai–Kim horizon thermodynamic derivation [Jacobson, 1995, Cai and Kim, 2005] be complete: every bit of classical information produced by an irreversible quantum-to-classical transition must be encoded somewhere. The only available surface at cosmological

scales is the cosmic horizon.

The sector separation follows immediately. Photons travel null geodesics: $d\tau = 0$. They do not accumulate proper time, do not participate in gravitational decoherence, and do not write to the cosmic ledger. They remain in the potential sector indefinitely. Massive particles travel timelike worldlines, participate in gravitational decoherence, and pay the Landauer cost at each virialization event. They are in the matter sector [Mahaffey, 2026f].

This sector separation is not imposed. It is derived from the geometry of the manifold combined with the thermodynamic requirement that classical information production carry a cost. It should be noted that local environmental decoherence — from lab walls, air molecules, or phonon coupling — is not a separate mechanism that bypasses the cosmic ledger. Every irreversible quantum-to-classical transition, local or cosmic, writes to the horizon. Local environments are relays, not substitutes: they mediate the irreversible interaction that produces the information, but the information itself is encoded on the cosmic horizon as the ultimate boundary of classical existence. The sector split governs which systems participate in this process at all, not where the information is first absorbed.

3 The Derivation Chain: From Jacobson to the Cosmic Ledger

The derivation chain is explicit and traceable. It proceeds in five steps.

Step 1: Jacobson (1995). The Einstein field equations are an equation of state derived from the proportionality of entropy and area for local causal horizons [Jacobson, 1995]. Gravity is thermodynamic. This derivation assumes the entropy functional contains only geometric terms.

Step 2: Cai and Kim (2005). The Friedmann equations follow from the same thermodynamic argument applied to the FRW apparent horizon [Cai and Kim, 2005]. Both derivations assume the entropy functional is complete.

Step 3: The missing contribution. Gravitational structure formation is the dominant source of irreversible classical information production in the universe. Each virialization event converts quantum superpositions to classical outcomes. The virial theorem for $1/r$ potentials partitions the gravitational binding energy exactly:

$$2K = |V|, \quad K = \frac{1}{2}|V|. \quad (1)$$

The kinetic half K is the decoherence energy — the Landauer cost of each quantum-to-classical transition. This identification carries a thermodynamic cost of $k_B T_H \ln 2$ per bit, where $T_H = \hbar H / 2\pi k_B$ is the Gibbons–Hawking temperature [Gibbons and Hawking, 1977]. The information must be encoded on the cosmic horizon.

Step 4: The accumulated ledger. Integrating the information production rate across 13.8 billion years of gravitational structure formation yields the activation function:

$$\mathcal{E}(a) = \exp\left(1 - \frac{1}{a}\right), \quad (2)$$

derived from the requirement that the cumulative informational entropy be consistent with the Jacobson–Cai–Kim horizon thermodynamics. This function is monotonically increasing by construction ($d\mathcal{E}/da > 0$ always), begins at the electroweak transition

($\mathcal{E} \rightarrow 0$ as $a \rightarrow 0$), equals unity today ($\mathcal{E}(1) = 1$), and asymptotes to e without arriving ($\mathcal{E}(\infty) = e$) [Mahaffey, 2026f].

Step 5: The cosmological consequence. Including the informational entropy contribution in the Cai-Kim first law modifies the matter perturbation equation through an additional Hubble friction term. The result is the dual-sector modification:

$$\mu(a) = \frac{H_{\Lambda\text{CDM}}^2}{H_{\Lambda\text{CDM}}^2 + \beta_m \mathcal{E}(a)} < 1, \quad \Sigma(a) = 1, \quad (3)$$

where $\beta_m = \Omega_m/2$ is derived from the virial theorem with zero free parameters [Mahaffey, 2026e]. The Planck 2018 MCMC posterior returns $\beta_m = 0.1583 \pm 0.0033$, consistent with the virial prediction at 0.2σ .

The derivation is complete. Every step traces from an established result. Nothing is postulated beyond what Jacobson and Cai-Kim already assumed, plus the requirement that the entropy budget be complete.

4 What Non-Locality Actually Is

With the ontological distinction established and the derivation chain explicit, the nature of quantum non-locality becomes clear.

Entangled systems are fields of potential that have not yet paid the Landauer cost of actualization. They have no classical identity, no definite position, no trajectory. They are not particles separated by a distance. They are one undivided field of potential that has not yet been partitioned into classical outcomes by an irreversible thermodynamic event.

Non-locality is the expected property of potential. Potential is not local. Locality is a property of actual things — classical particles with definite positions on timelike worldlines. Before actualization, the concept of spatial separation does not apply in the classical sense, because there is no classical object to be separated.

When measurement occurs — when an irreversible Landauer transaction is forced on a timelike worldline — the field of potential actualizes. At that moment it becomes a particle, writes to the cosmic ledger, and locality is restored. There is no mechanism required for the apparent instantaneous correlation because there is no signal traveling between two separated particles. There is one field of potential becoming classical at one end, and the entanglement resolving as a consequence.

Bell’s theorem [Bell, 1964] is precisely correct: there are no local hidden variables. IAM agrees, but for a derived reason rather than a postulated one. There are no hidden variables because there is no classical particle to have hidden variables until actualization occurs. Bell’s result is exactly what one would predict from IAM’s ontology.

The sector assignment makes a quantitative prediction. Photon-based entangled systems ($\Sigma = 1$) do not decohere during transit. Their entanglement is preserved at arbitrary distances, and the CHSH S -parameter remains $|S| = 2\sqrt{2}$ identically, in full agreement with quantum mechanics. Matter-based entangled systems ($\mu < 1$) experience gravitational decoherence on timelike worldlines. The Bell violation decays as [Mahaffey, 2026a]:

$$S(t) = 2\sqrt{2}(1 - D(t)), \quad D(t) = \mathcal{E}(t/\tau_{\text{IAM}})/e, \quad (4)$$

where τ_{IAM} is the gravitational decoherence timescale, derived from first principles [Mahaffey, 2026c]:

$$\tau_{\text{IAM}} = \frac{\hbar k_B^2 T^2 \ln 2}{E_G^3}, \quad (5)$$

where $E_G = Gm^2/r$ is the gravitational self-energy of the superposition and T is the bath temperature. This formula contains no free parameters. Table 1 gives representative values across three experimental regimes and compares IAM against Penrose–Diósi and standard environmental decoherence.

Table 1: Gravitational decoherence timescales across three experimental regimes. IAM predicts a ramp profile with T^2 temperature scaling and m^{-6} mass scaling. Penrose–Diósi predicts exponential decay with no temperature dependence and $m^{-5/3}$ scaling. The mesoscopic frontier ($\sim 10^{-12}$ kg) is where IAM and Penrose–Diósi predictions diverge measurably with current experimental reach.

System	IAM τ_{IAM}	P–D τ_{PD}	Key discriminator
Trapped ion ($\sim 10^{-25}$ kg)	$\gg$ practical	$\gg$ practical	Both irrelevant
Superconducting qubit ($\sim 10^{-20}$ kg)	$\gg$ practical	$\gg$ practical	Both irrelevant
Nanosphere (10^{-12} kg, 10 mK)	$\approx 560 \mu\text{s}$	$\sim 10 \mu\text{s}$	Profile shape; T^2 vs T^0

The photon-matter asymmetry — photon entanglement maintained at 1,200 km (Micius satellite, 2017 [Yin et al., 2017]) versus matter entanglement limited to approximately 1.3 m (trapped ions, 2019 [Hensen et al., 2015]) — is a ratio of one million, consistent with the sector separation.¹

A clarification on massive unstable particles is warranted. Entanglement has been confirmed between top quark pairs at the Large Hadron Collider [CMS Collaboration, 2024], despite top quarks being massive fermions on timelike worldlines. This is consistent with IAM. Gravitational decoherence operates on virialization timescales — the timescale for gravitational collapse to convert quantum superpositions to classical outcomes, which is orders of magnitude longer than the top quark lifetime of $\sim 10^{-25}$ s. Top quarks decay before gravitational decoherence can act. The IAM prediction is that entanglement between massive systems decays on *gravitational* decoherence timescales, not on particle decay timescales. The two mechanisms are distinct.

A further open question is entanglement across time rather than space. Experiments have demonstrated correlations between particles that never coexisted simultaneously [Megidish et al., 2013], suggesting entanglement is a property of the quantum state rather than of spatially separated objects. In IAM, the field of potential exists prior to actualization regardless of whether the actualization events are spacelike or timelike separated. The temporal case is not excluded by the framework, but a full treatment of temporal entanglement within IAM is beyond the scope of this paper and is identified as an open direction.

¹Recent results have demonstrated Bell inequality violations via quantum indistinguishability through path identity, without requiring entanglement in the conventional sense [Zhang et al., 2025]. This does not challenge the IAM framework: such violations arise from indistinguishability structure, not from entangled fields of potential in the IAM sense. The sector-dependent decoherence prediction applies specifically to systems whose correlations originate from a shared quantum state, not from post-selection on indistinguishable paths.

5 The Bridge: $n = 5/2$ from Quantum to Cosmic

The claim that a single thermodynamic mechanism operates from the quantum boundary to the cosmological scale requires a bridge. That bridge is the critical exponent $n = 5/2$, derived independently from two directions.

5.1 Bottom-Up: From Quantum Decoherence Physics

The Joos–Zeh result establishes that gravitational decoherence of macroscopic collapsed structures is instantaneous on cosmological timescales [Joos and Zeh, 1985]. The rate-limiting step is therefore virialization, not decoherence itself. Integrating the information production rate over the Press–Schechter halo mass function [Press and Schechter, 1974], with peak height scaling $\nu \propto D(a)^{-1/2}$ for halos near the characteristic mass M_* :

$$\dot{I} \propto \rho_m(a) \cdot D(a)^{5/2} \cdot f(a) \cdot H(a), \quad (6)$$

with $n = 5/2$ derived from quantum Darwinism applied to gravitational structure formation [Mahaffey, 2026b].

5.2 Top-Down: From Horizon Thermodynamic Consistency

The top-down derivation requires the cumulative informational entropy to reproduce $\mathcal{E}(a) = \exp(1 - 1/a)$. Substituting matter-domination scalings and writing $\dot{I} \propto \rho_m D^n H$ (the derivation is presented under matter-domination as the epoch during which the bulk of structure formation occurs; the full treatment across all epochs is given in [Mahaffey, 2026b]):

$$S_{\text{info}}(a) \propto \int a^{n-9/2-1} da \propto a^{n-9/2}. \quad (7)$$

Requiring $S_{\text{info}}(a) \propto a^{-1}$ (from the $\exp(1 - 1/a)$ functional form) forces:

$$n - \frac{9}{2} = -1 \quad \implies \quad \boxed{n = \frac{5}{2}}. \quad (8)$$

5.3 Convergence

Both directions yield $n = 5/2$. The bottom-up derivation obtains this from quantum decoherence physics integrated over the Press–Schechter mass function. The top-down derivation obtains it from requiring the activation function to be consistent with horizon thermodynamics. Their convergence is not a coincidence. It constitutes a derivation connecting the quantum boundary of actualization to cosmological observables from two independent directions simultaneously [Mahaffey, 2026b].

The same irreversible thermodynamic process that resolves entanglement at the quantum scale accumulates into the ledger $\mathcal{E}(a)$ that suppresses structure growth at the cosmological scale. One mechanism. One law.

6 IAM’s Law

The results of the preceding sections can be stated as a single thermodynamic law:

IAM’s Law. Every irreversible crossing from potential to actual — every quantum-to-classical transition on a timelike worldline — pays a thermodynamic cost of $k_B T_H \ln 2$ per bit to the cosmic horizon. This cost is physically real, thermodynamically irreversible, and cannot be recovered. Classical existence is continuously purchased, one decoherence event at a time.

This law has three direct consequences:

1. Particles are post-actualization concepts. Before the Landauer cost is paid, there is no particle. There is a field of potential. The word “particle” applied to a pre-measurement quantum system is a category error.

2. Non-locality is the expected behavior of potential. Entangled systems are fields of potential that have not paid the cost of actualization. Non-locality is not a mystery to be explained — it is the natural state of potential before the boundary is crossed.

3. The arrow of time is a thermodynamic necessity. The cosmic ledger $\mathcal{E}(a)$ is monotonically increasing because each Landauer transaction is irreversible. Time has direction because actualization has direction. The ledger only grows [Mahaffey, 2026f].

7 Observational Confirmation

IAM’s law is not a philosophical claim. It is a physical claim with measured consequences.

Seventeen MCMC chains were run against the full Planck 2018 likelihood (TT, TE, EE + low-ell + lensing). All 17 converged to $R - 1 < 0.01$ with no discarded runs, no parameter adjustments, and no excluded redshift ranges. The results [Mahaffey, 2026d]:

- $\sigma_8 = 0.7998 \pm 0.0058$ (suppressed from Λ CDM’s 0.809, consistent with KiDS-Legacy, DES Y3, HSC Y3 at 0.1σ)
- $H_0^{(\text{photon})} = 67.16 \pm 0.47 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (0.37σ from Planck)
- $H_0^{(\text{matter})} = 72.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (0.75σ from SH0ES)
- $\beta_m = 0.1583 \pm 0.0033$ (0.2σ from virial prediction 0.1575)
- $\Delta\chi^2 = +0.54$ best-fit vs Λ CDM

The sector-dependent Hubble split — photon-sector observables on null geodesics vs matter-sector observables on timelike worldlines — is a direct consequence of IAM’s law. The two “values” of H_0 are not a tension. They are measurements of two distinct physical quantities separated by the accumulated Landauer cost of 13.8 billion years of actualization.

The predicted matter-sector modification today is $\mu_0 - 1 = -0.136$, derived from the virial theorem coupling $\beta_m = \Omega_m/2$ and the activation function $\mathcal{E}(a)$, with zero free parameters beyond Λ CDM. The existing chain results are already consistent with this value. The Euclid DR1 measurement (October 2026), with projected sensitivity $\sigma(\mu_0) \approx 0.04$, will determine the precision of the mechanism’s magnitude — not whether the mechanism operates, which the Planck 2018 chains have already established.

8 Discussion

The framework presented here does not modify quantum mechanics. It identifies what quantum mechanics has always been describing: the behavior of potential before it crosses the thermodynamic boundary into classical existence.

Standard quantum mechanics is the correct theory of potential. General relativity is the correct theory of actual spacetime geometry. IAM is the thermodynamic account of the boundary between them — the cost of crossing, the ledger of crossings, and the cosmological consequence of 13.8 billion years of such crossings accumulated on the cosmic horizon.

The measurement problem, the arrow of time, quantum non-locality, and the Hubble and σ_8 tensions are not separate problems. They are different aspects of one problem: the incomplete entropy budget of the Jacobson–Cai–Kim derivation. Including the Landauer cost of irreversible bulk information production closes the budget, addresses the foundational paradoxes, and produces testable cosmological predictions.

The fact that decoherence is thermodynamically irreversible has been accepted since the 1980s [Joos and Zeh, 1985, Zurek, 1981]. The fact that horizons carry entropy has been accepted since Bekenstein [Bekenstein, 1973] and Hawking [Hawking, 1975]. The fact that gravity is thermodynamic has been accepted since Jacobson [Jacobson, 1995]. The missing step was asking where the Landauer cost of every irreversible quantum-to-classical transition goes at cosmological scales, and what happens when 13.8 billion years of such costs accumulate.

The answer is $\mathcal{E}(a)$. The consequences are $\mu < 1$ and $\Sigma = 1$. The confirmation is 17 converged chains against the full Planck 2018 likelihood.

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The thermodynamic framework of T. Jacobson [Jacobson, 1995] is the foundation upon which this work is built. His 1995 derivation of the Einstein equations as an equation of state established that gravity is thermodynamic; IAM is the completion of that derivation when the entropy budget is made explicit. The quantum Darwinism framework of W.H. Zurek [Zurek, 2003, 2009] provided the local account of the quantum-to-classical transition that this paper extends to cosmological scales. The thermal time program of C. Rovelli [Connes and Rovelli, 1994] posed the foundational question of the thermodynamic origin of time that the activation function $\mathcal{E}(a)$ addresses. A personal response from Professor P.J.E. Peebles identified the quantum-to-classical derivation as the foundational gap requiring closure before the cosmological framework could be trusted — that identification directly motivated the bridge from quantum decoherence physics to the cosmic ledger presented in Section 5.

Data Availability

All MCMC chains, analysis scripts, and prediction figures are publicly archived at <https://github.com/hmahaffeyges/IAM-Validation> with canonical Zenodo DOI [10.5281/zenodo.18702042](https://doi.org/10.5281/zenodo.18702042). Repository timestamps confirm all predictions were established prior to any observational comparison.

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